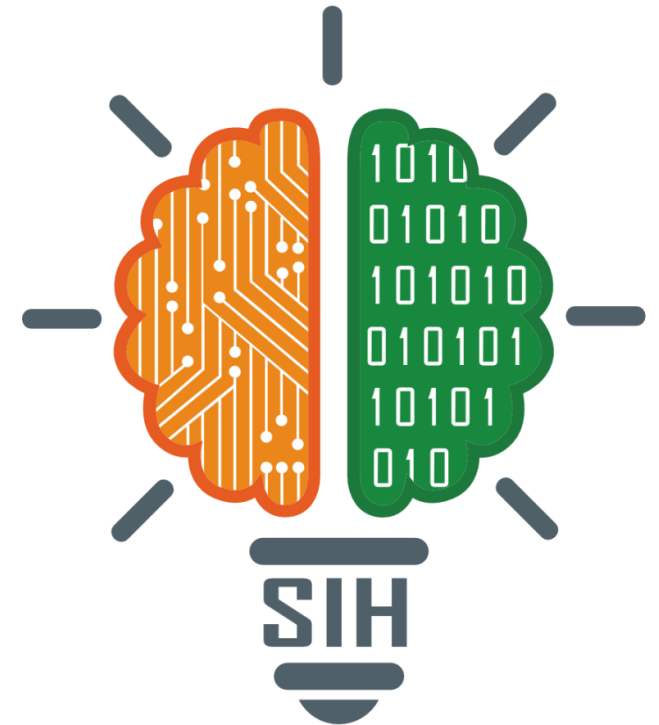


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID – 26034**
- **Problem Statement Title - Software System to check compliance of Packaged Commodities under Legal Metrology**
- **Theme- Miscellaneous**
- **PS Category- Software**
- **Team ID-**
- **Team Name - LabelSentinals**





Capture

AI / OCR
ExtractionRule
ValidationViolation
DetectionInspector
VerificationAutomated
Report

From Package Image to Evidence-Backed Legal Metrology Compliance

DETAILED
SOLUTION

- **AI-POWERED CAPTURE:** Uses smartphone photos/video with Computer Vision + OCR to read key package declarations.
- **STRUCTURED EXTRACTION:** Converts MRP, net quantity, dates, manufacturer/packer/importer and other fields into structured data.
- **INSPECTION WORKFLOW:** Links every finding to the inspection case and captured evidence.

HOW IT
ADDRESSES
THE PROBLEM

- **FASTER INSPECTION:** Reduces repetitive manual reading and checking of package labels.
- **RULE-BASED VALIDATION:** A category-aware Legal Metrology Rule Engine checks applicable requirements.
- **EVIDENCE + VERIFICATION:** Highlights the relevant image region; confidence scores let inspectors approve or override findings.

INNOVATION &
UNIQUENESS

- **NOT JUST OCR:** Combines Computer Vision, OCR, information extraction and legal-rule validation.
- **EVIDENCE-BACKED FINDINGS:** Every potential violation is linked to visual evidence for review.
- **HUMAN-IN-THE-LOOP:** Inspector remains the final decision-maker before report generation.

OUTPUT: COMPLIANCE STATUS • VIOLATION EVIDENCE • PDF / EDITABLE INSPECTION REPORT

1. FRONTEND UI
(Progressive Web App)**React 19**
(TypeScript)**Vite****Tailwind CSS v4****Lucide React**
(Scalable Icons)

- Capture Label
- Crop & Preview
- View Results
- Generate Report

**Progressive
Web App (PWA)****2. BACKEND API
& LOGIC****FastAPI**

(Python)



- High-performance
- Asynchronous Endpoints
- RESTful API
- Authentication & Authorization

**3. DATABASE &
ORM****PostgreSQL****SQLAlchemy**

(ORM)

**Alembic**
(Migrations)

- Relational Database
- ACID Compliance
- Secure & Scalable

**4. AI & COMPUTER
VISION CORE****PaddleOCR**

(PaddlePaddle)

Robust Text Extraction
from Complex Labels**Google
GenAI**

(Gemini)

Legal Metrology
Rule Engine &
NLP Validation

Rule Engine



NLP Validation

**5. DOCUMENT
GENERATION****WEASYPRINT**
(HTML to PDF)**FPDF2**

(PDF Generation)



- Compliance Report
- Evidence Attachments
- Audit Ready
- Digital Signature Support

**IMPLEMENTATION PROCESS****1 IMAGE ACQUISITION &
PREPROCESSING**

The PWA captures the product label image. **react-easy-crop** allows the inspector to focus on relevant label areas before the backend processes it.

**2 OCR & TEXT EXTRACTION**

PaddleOCR analyzes the image to accurately extract raw text, handling different fonts, rotations, and complex packaging layouts.

**3 AI RULE ENGINE VALIDATION**

The raw text is passed to the Generative AI engine. The engine acts as a "Legal Metrology Expert," structuring the data and mapping it against **mandatory FSSAI/** Metrology rules.

**4 INSPECTOR VERIFICATION
(HUMAN-IN-THE-LOOP)**

The extracted data and compliance scores are sent back to the React UI. The inspector reviews the AI's findings, visually verifying the flagged evidence.

**5 FINALIZATION & REPORTING**

Once verified, the compliance status is securely saved to PostgreSQL, and a formal, evidence-backed PDF report is generated for auditing.

TECHNOLOGIES TO BE USEDReact 19
(TypeScript)

Vite



Vite

Tailwind
CSS v4Lucide
React

Python



FastAPI



PostgreSQL

SQLAlchemy

Genjlmy



Alembic



PaddleOCR

Google
GenAI

WeasyPrint



FPDF2

Feasibility

- **Technically highly feasible** with existing AI technologies like PaddleOCR and GenAI.
- **Operationally simple** using standard smartphone cameras, ensuring immediate usability for inspectors and QA teams.
- **Economically viable** by automating manual inspection, reducing overhead and scaling cost-effectively on cloud.

Viability

- User friendly interface and **dashboards** drive both farmer and official engagement.
- Farm based **rewards** and community recognition boost participation in compliance and reporting.
- Active participation in **crop switching** initiatives can be incentivized with reward points.

Practical Implementation

- MVP ready for **pilot rollout in real farms** within **6 to 9 months**.
- Works **offline** for remote rural areas with seamless data sync.
- Simple **multilingual** app and UI for effortless access by all farmers.

Potential Challenges and Risks

- 01 Image Quality & Variations:**
Poor lighting, blurriness, or curved/reflective packaging can cause OCR failures.
- 02 Non-Standardized Layouts:**
Food labels have massive variations in design, making template-matching impossible.
- 03 OCR Accuracy Limitations:**
Complex fonts, clustered text, or local languages may lead to misread information.
- 04 Regulatory Changes:**
Frequent updates to Legal Metrology rules could make hardcoded compliance logic obsolete.
- 05 Connectivity Issues:**
Patchy internet in rural areas may disrupt real-time data sync and updates.

Pre-processing & Feedback:

Allow cropping with react-easy-crop and enhance images before OCR for better results.

AI vs Templates:

Use a Generative AI Rule Engine to parse unstructured text and identify data regardless of layout.

Human-in-the-Loop:

Show AI confidence scores and require final inspector verification to ensure accuracy.

Centralized Rule Engine:

Rules managed in backend (FastAPI/PostgreSQL) enabling instant updates to regulations.

Offline-First & Sync:

Enable offline capture and local storage with auto sync once internet is available.

Strategies For Overcoming Challenges

01

02

03

04

05

POTENTIAL IMPACT ON TARGET AUDIENCE

**Field Officers & Inspectors**

10× faster audits: 25–30 min → under 4 sec.
Automated OCR + AI checks 12 mandatory Rule 6 declarations.
Instant Form-1 memos, evidence photos & audit trails.

**Consumers & Citizens**

Protects MRP, quantity and consumer-care declarations.
Supports public verification, photo complaints and real-time docket tracking.

**Manufacturers & Retailers**

Creates a level playing field against deceptive packaging and duplication.
Pre-market screening against Legal Metrology & FSSAI requirements.

KEY SOLUTION BENEFITS

**Faster Enforcement**

Rapid label reading, automated checks and digital evidence reduce inspection effort.

**Transparency & Trust**

Signed officer verification, traceable records and evidence-backed decisions improve accountability.

**Multilingual Access**

Detects package other regional language aging text across English, Hindi, Telugu .

**Paperless Enforcement**

Digital dossiers, inspection histories and PDF reports reduce paperwork and storage.

ECONOMIC • SOCIAL • ENVIRONMENTAL

**Economic Benefits**

- Curbs illegal MRP markups & quantity deception.
- Speeds identification of non-compliance and penalties.
- Up to 90% lower administrative effort through digitisation.

**Social & Consumer Welfare**

- Improves confidence in price, quantity, dates and licences.
- Supports regional-language enforcement.
- AI cues + mandatory officer verification promote fair decisions.

**Environmental & Operational**

- Paperless inspection chain and cloud dossiers.
- Less physical evidence storage.
- Efficient image/OCR processing reduces unnecessary bandwidth and compute.

RESULT: Faster inspections • Stronger compliance • Better consumer protection • Digital, accountable enforcement



LegalMetriX Research Application

1. 360° video capture → automatic keyframe extraction
2. Variance of Laplacian → sharpness filtering
3. PP-OCRv4 → multilingual text detection and recognition
4. Multimodal AI → entity extraction and statutory rule cross-checking
5. Compliance engine → evidence-backed violations and reporting

AI, Computer Vision & Multimodal Research Papers

[PP-OCRv4 — Du et al. \(2023\)](#)

DBNet text detection + SVTR recognition for multilingual packaging text extraction.

[Gemini: A Family of Highly Capable Multimodal Models \(2023\)](#)

Cross-referencing detected text with statutory rules, dual-MRP detection and bounding-box localisation.

[Modified Laplacian — Pech-Pacheco et al. \(2000\)](#)

Variance of Laplacian sharpness filtering to reject motion blur and select sharp package panels.

Technical Implementation & Project Artifacts

[Live Deployed Application — Click Here to view](#)

[Platform Source Code \(GitHub\) — Click Here to view](#)

[National Open Packaging Corpus — Open Food Facts India Dataset](#)

[Retail Label Multi-Oriented Text Benchmark — Click Here to view](#)

Regulatory Compliance Context

[Consumer Protection \(E-Commerce\) Rules, 2020](#)

Supports digital marketplace compliance for pre-packaged commodities and consumer-facing product information.

[Ministry of Consumer Affairs, Food & Public Distribution](#)

[Food Safety and Standards Authority of India \(FSSAI\)](#)

Quick Reference Links

[Legal Metrology Portal](#)

[FSSAI Official Website](#)

[PP-OCRv4 Research Paper \(arXiv\)](#)

[Gemini Research Paper \(arXiv\)](#)

[LegalMetriX Live Application](#)

[LegalMetriX GitHub Repository](#)